Baking kamacite for arsenic

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Abstract

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1 Elements and their vapor pressures

The table below shows a list of elements and temperatures at which they achieve the corresponding vapor pressure. The temperatures are given in kelvin. These values had been sourced from wikipedia pages for each of these elements.

atomic number	element name	1Pa	10Pa	100Pa	1kPa	10kPa	100kPa
26	iron	1728	1890	2091	2346	2679	3132
27	cobalt	1790	1960	2165	2423	2755	3198
28	nickel	1783	1950	2154	2410	2741	3184
29	copper	1509	1661	1850	2089	2404	2834
30	**zinc**	610	670	750	852	990	1179
31	gallium	1310	1448	1620	1838	2125	2518
32	germanium	1644	1814	2023	2287	2633	3104
33	**arsenic**	553	596	646	706	781	874
34	**selenium**	500	552	617	704	813	958
42	molybdenum	2742	2994	3312	3707	4212	4879
44	ruthenium	2588	2811	3087	3424	3845	4388
45	rhodium	2288	2496	2749	3063	3405	3997
46	palladium	1721	1897	2117	2395	2753	3234
47	silver	1283	1413	1575	1782	2055	2433
48	**cadmium**	530	583	654	745	867	1040
49	indium	1196	1325	1485	1690	1962	2340
50	tin	1497	1657	1855	2107	2438	2893
51	**antimony**	807	876	1011	1219	1491	1858
52	**tellurium**	?	?	(775) ?	(888) ?	1042	1266
74	tungsten	3477	3773	4137	4579	5127	5823
75	rhenium	3303	3614	4009	4500	5127	5954
76	osmium	3160	3423	3751	4148	4638	5256
77	iridium	2713	2957	3252	3614	4069	4659
78	platinum	2330	(2550) ?	2815	3143	3556	4094
79	gold	1646	1814	2021	2281	2620	3078
80	**mercury**	315	350	393	449	523	629
81	**thallium**	882	977	1097	1252	1461	1758
82	**lead**	978	1088	1229	1412	1660	2027
83	**bismuth**	941	1041	1165	1325	1538	1835

Elements which have vapor pressures of 1Pa below 1000 kelvin had been marked in bold, and they are as follows: Zinc, Arsenic, Selenium, Cadmium, Antimony, Tellurium, Mercury, Thallium, Lead, Bismuth.

Additionally, elements which have vapor pressures of 1Pa at higher than 1000K but at least 200K lower than iron (1528K) had been underlined, and they are as follows: Copper, Gallium, Silver, Indium, Tin.

2 original composition

The table below shows a breakdown of elements constituting the kamacite being processed. "Normalized kamacite mass fraction" is the mass of the element divided by the original mass of kamacite, that is to say, for example with iron, given that this figure is given as "0.94043381955614200000" for iron, it would mean that every 1kg of kamacite contains 0.940433819556142kg of iron.

atomic number	element name	normalized kamacite mass fraction
26	iron	0.94043381955614200000
27	cobalt	0.00252207251608238000
28	nickel	0.05557108933740840000
29	copper	0.00047021690977807100
30	zinc	0.00076944585236411600
31	gallium	0.00003248771376648490
32	germanium	0.00008976868277581350
33	arsenic	0.00000769445852364116
34	selenium	0.00005557108933740840
42	molybdenum	0.00000512963901576077
44	ruthenium	0.00000346250633563852
45	rhodium	0.00000068395186876810
46	palladium	0.00000282130145866843
47	silver	0.00000059845788517209
48	cadmium	0.00000188086763911228
49	indium	0.00000018808676391123
50	tin	0.00000512963901576077
51	antimony	0.00000051296390157608
52	tellurium	0.00000897686827758135
74	tungsten	0.00000051296390157608
75	rhenium	0.00000020946025981023
76	osmium	0.00000282130145866843
77	iridium	0.00000230833755709235
78	platinum	0.00000418920519620463
79	gold	0.00000072669886056611
80	mercury	0.00000106867479495016
81	thallium	0.00000033342653602445
82	lead	0.00000598457885172090
83	bismuth	0.00000029495424340624

3 expected extraction results

"actual amount extracted" is the total mass of that element in kilograms which could be pulled out of one kilogram of kamacite.

"normalized product mass fraction" is the mass of that element divided by the total amount of extracted material.

For 1000K extraction, the total amount of mass extracted from one kilogram of kamacite is 0.000851468780226131kg (851.4687802 milligrams).
Majority of the product (over 90% by mass) is zinc.

atomic number	element name	actual amount extracted	Normalized product mass fraction
30	zinc	0.0007694458523641160	0.90366889571660900
33	arsenic	0.0000076944585236412	0.00903668895716609
34	selenium	0.0000555710893374084	0.06526497580175520
48	cadmium	0.0000018808676391123	0.00220896841175171
51	antimony	0.0000005129639015761	0.00060244593047774
52	tellurium	0.0000089768682775814	0.01054280378336040
80	mercury	0.0000010686747949502	0.00125509568849529
81	thallium	0.0000003334265360245	0.00039158985481053
82	lead	0.0000059845788517209	0.00702853585557363

For 1500K extraction the total mass extracted from one kilogram of kamacite is 0.00136008958743553kg (1.360089587 grams). Majority of the product is zinc (56.57% by mass) followed by copper (34.57% by mass).

atomic number	element name	actual amount extracted	Normalized product mass fraction
29	copper	0.0004702169097780710	0.345724953798574
30	zinc	0.0007694458523641160	0.565731742579485
31	gallium	0.0000324877137664849	0.023886451353356
33	arsenic	0.0000076944585236412	0.00565731742579485
34	selenium	0.0000555710893374084	0.0408584036307406
47	silver	0.0000005984578851721	0.000440013577561822
48	cadmium	0.0000018808676391123	0.00138289981519429
49	indium	0.0000001880867639112	0.000138289981519429
50	tin	0.0000051296390157608	0.0037715449505299
51	antimony	0.0000005129639015761	0.00037715449505299
52	tellurium	0.0000089768682775814	0.00660020366342733
80	mercury	0.0000010686747949502	0.000785738531360395
81	thallium	0.0000003334265360245	0.000245150421784443
82	lead	0.0000059845788517209	0.00440013577561822

Both temperatures would get rid of the original majority elements Iron (94%) and Nickel (5.56%), reducing the amount of material that will be passed to downstream processes by two orders of magnitude.

4 Implications on calutron extraction

Below is a table showing the energy needed to ionize a given element as a function of mass:

atomic number	element name	specific ionization energy (MJ/kg)
29	copper	11.7316589557171
30	zinc	13.8635668400122
31	gallium	8.3014213387261
33	arsenic	12.6398803976344
34	selenium	11.9157665472135
47	silver	6.77678871066728
48	cadmium	7.7196790435355
49	indium	4.8624780086746
50	tin	5.96916856204195
51	antimony	6.84954007884363
52	tellurium	6.81269592476489
80	mercury	5.02068896754574
81	thallium	2.88384382033467
82	lead	3.45366795366795

To get the specific ionization energy of the two extraction results, a weighted average is taken of the specific ionization energies, with the mass fraction of the element in the mix being the weight.

For the 1000K extraction product, the result is 8.47949943193412MJ/kg. The majority of cost comes from zinc.

atomic number	element name	specific ionization energy (MJ/kg)	1000K fraction	weighted ion. energy (MJ/kg)
30	zinc	13.8635668400122	0.565731742579485	7.84305982676727
33	arsenic	12.6398803976344	0.00565731742579485	0.0715078156334998
34	selenium	11.9157665472135	0.0408584036307406	0.486859199155726
48	cadmium	7.7196790435355	0.00138289981519429	0.0106755427226645
51	antimony	6.84954007884363	0.00037715449505299	0.00258333482978149
52	tellurium	6.81269592476489	0.00660020366342733	0.0449651806004497
80	mercury	5.02068896754574	0.000785738531360395	0.00394494877577673
81	thallium	2.88384382033467	0.000245150421784443	0.000706975528915504
82	lead	3.45366795366795	0.00440013577561822	0.0151966079200405
			sum (MJ/kg)	**8.47949943193412**

For the 1500K extraction product, the result is 12.7598854779392MJ/kg. The majority of energy cost is added by zinc and copper.

atomic number	element name	specific ionization energy (MJ/kg)	1000K fraction	weighted ion. energy (MJ/kg)
29	copper	11.7316589557171	0.345724953798574	4.05592725044592
30	zinc	13.8635668400122	0.565731742579485	7.84305982676727
31	gallium	8.3014213387261	0.023886451353356	0.198291496971192
33	arsenic	12.6398803976344	0.00565731742579485	0.0715078156334998
34	selenium	11.9157665472135	0.0408584036307406	0.486859199155726
47	silver	6.77678871066728	0.000440013577561822	0.00298187904496128
48	cadmium	7.7196790435355	0.00138289981519429	0.0106755427226645
49	indium	4.8624780086746	0.000138289981519429	0.00067243199395824
50	tin	5.96916856204195	0.0037715449505299	0.0225129875490311
51	antimony	6.84954007884363	0.00037715449505299	0.00258333482978149
52	tellurium	6.81269592476489	0.00660020366342733	0.0449651806004497
80	mercury	5.02068896754574	0.000785738531360395	0.00394494877577673
81	thallium	2.88384382033467	0.000245150421784443	0.000706975528915504
82	lead	3.45366795366795	0.00440013577561822	0.0151966079200405
			sum (MJ/kg)	**12.7598854779392**